Natnael Alemseged Astaw

Applied AI Engineer | Agent Systems, Evaluation & Production AI

Addis Ababa, Ethiopia | +251 961 261 683 | natiaabaydam@gmail.com
Portfolio | LinkedIn | GitHub | Upwork

PROFILE

Software engineer with 4+ years of experience shipping production AI, web, and mobile products. Builds agentic systems, RAG, evaluation and fine-tuning pipelines, LLM-powered APIs, and polished cross-platform applications using Python, TypeScript, Flutter, and AWS. Experienced in taking ambiguous product requirements from discovery through deployment, observability, and iteration.

EXPERIENCE

Senior Software Engineer
Jul 2025 - Present

Hire Armada | United States (Remote)

- Engineer AI-driven backends and product features with Node.js, FastAPI, Next.js, Flutter, Prisma, LangGraph, MCP, Composio, and vector memory.
- Built a social discovery platform backend supporting 10K+ daily active users, real-time matching and messaging, and 99.9% uptime.
- Shipped a personalized affirmation product with Gemini chat, home-screen widgets, and adaptive feeds, increasing retention by 35%.
- Delivered an agentic collaboration workspace with real-time events, email automation, semantic search, and tagged agents, improving team productivity by 20%.

Senior Full-Stack Developer
Jan 2025 - May 2026

Startup Agile | United States (Remote)

- Led Next.js, React, React Native, and Expo delivery across web and mobile products; completed milestones 15% ahead of schedule.
- Implemented OTA updates, deep linking, Firebase notifications, Google Maps, AI assistants, and product analytics while reducing mobile load time by 35%.
- Improved Next.js rendering, SEO, and Core Web Vitals, contributing to 50%+ web traffic growth and a 30% performance improvement.
- Standardized TypeScript, tRPC, Prisma, Playwright, and Detox workflows, reducing production defects by 65%.

AI Engineer
Jan 2025 - Aug 2025

DataCore Software | United States (Remote, Contract)

- Built FastAPI services for real-time detection, recognition, transcription, summarization, and translation using self-hosted LLMs.
- Deployed containerized inference workloads to AWS EKS with sub-200 ms inference and 99.9% uptime at peak load.
- Fine-tuned open models and optimized GPU inference with TensorRT and ECR images, reducing cost by 12%; automated delivery with Terraform and GitHub Actions.

Senior Mobile Developer
Aug 2024 - Sep 2025

Qemer Software Technology PLC | Addis Ababa, Ethiopia (On-site)

- Led delivery of 10+ Flutter and Ionic applications, accelerating release cycles by 40%.
- Improved application architecture and runtime performance, cutting load times by up to 50%.
- Mentored junior engineers and established reusable delivery patterns, reducing onboarding time by 30%.

EARLIER EXPERIENCE

ERP Functional Consultant
Mar 2023 - Sep 2024

Red Cloud ICT Solutions | Addis Ababa, Ethiopia

- Progressed from Junior Consultant to Functional Consultant Assistant, supporting ERP discovery, configuration, testing, user enablement, and implementation delivery.

Freelance Web Developer
Aug 2022 - Mar 2023

Independent | Remote

- Designed, built, and delivered responsive web solutions directly with clients from requirements through launch.

SELECTED INDEPENDENT DELIVERY

Social Music Party Game: Delivered a complete client product engagement from Jan-Jun 2026; received a verified 5.0 Upwork rating.

AI Chatbot Engineering: Completed a focused AI chatbot engagement in Jan 2026; received a verified 5.0 Upwork rating.

Flutter Inventory Tracker: Shipped and supported a cross-platform inventory application in 2025; received a verified 5.0 Upwork rating.

Upwork Top Rated | 100% Job Success | Verified 5-star client outcomes

TECHNICAL SKILLS

AI & Agents: LangGraph, RAG, agent orchestration, LLM-as-a-judge, statistical evaluation, fine-tuning, LoRA, DPO/SimPO, structured generation, zero-shot prompting, MCP, vector databases

ML Experimentation: Jupyter Notebook, Google Colab, reproducible experiments, benchmark design, error analysis, sampling and calibration analysis

Backend & Data: Python, FastAPI, Node.js, Express.js, TypeScript, PostgreSQL, MongoDB, Redis, Supabase, Firebase, Prisma, GraphQL, REST APIs, Socket.io

Web & Mobile: Next.js, React, React Native, Expo, Flutter, Dart, Ionic, Redux Toolkit, TanStack Query, BLoC

Cloud & Delivery: AWS, Docker, Kubernetes/EKS, Terraform, GitHub Actions, Prometheus, Grafana, CI/CD, microservices, n8n

SELECTED TECHNICAL WRITING

When Generic Benchmarks Fail: Building a Sales-Domain Evaluation Bench from Scratch
DEV Community | May 2026

Why Pairing Your Bootstrap Is Necessary, And When It Stops Helping
DEV Community | May 2026

CERTIFICATIONS

AI Agent Engineering, AI Evaluation & Forward-Deployed Engineering
10 Academy | Issued May 2026

AWS Certified Solutions Architect - Associate
AWS | Feb 2024 - Feb 2027

AWS Certified Cloud Practitioner
AWS | Oct 2023 - Feb 2027

Founders Academy
ALX Africa | Issued Jul 2024

CCNA: Introduction to Networks
Cisco | Issued Nov 2021

EDUCATION

Bachelor of Science in Information Systems
Jan 2018 - Dec 2022

Hawassa University | GPA: 3.57